Exact classification of ordered regular polygon Chambolle–Pock cycles

Abstract

We classify exact paired regular polygon cycles of the literal primal-first Chambolle–Pock map in the ordinary Euclidean plane. For every integer $m \geq 4$, positive extrapolation excludes the prescribed ordered regular m-gon orbit at every positive stepsize product. At zero extrapolation, $m \geq 5$ admits exactly one product and one dual vertex, whereas the square admits an explicit exceptional family. The proof establishes the equivalence of all vertex normal inequalities to four adjacent tests, including every boundary case. The zero-extrapolation minima are $4\sin^2(\pi/m)$, with unattained infimum zero over finite polygons. These are restricted-class statements, not a classification of arbitrary cycles or a universal convergence theorem. No novelty claim is made.

Keywords: Chambolle–Pock iteration; regular polygon cycles; convex projection; normal-cone inequalities; zero extrapolation; exact product classification

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1 The map and the precise polygon class

All inner products and norms are Euclidean on $\mathbb{R}^2$. We consider the primal-first recurrence associated with the Chambolle–Pock framework [1], with its order specified literally:

$$\begin{aligned} x_{n+1} &= P_C(x_n - r y_n), \\ y_{n+1} &= P_D\big(y_n + (1+\theta)x_{n+1} - \theta x_n\big), \end{aligned} \tag{1}$$

where $r > 0$ and $\theta \geq 0$. There is no outer relaxation or historical state beyond the current pair. In proximal notation, the data are

$$f = \iota_C, \quad g^* = \iota_D, \quad L = I, \quad \tau = r, \quad \sigma = 1.$$

Here ι_E is the indicator of a closed convex set E, and g is the support function of D. In particular, $\tau\sigma\|L\|^2 = r$.

Fix an integer $m \geq 4$ and define

$$\alpha = \frac{2\pi}{m}, \quad a = \cos\alpha, \quad b = \sin\alpha, \quad \delta = 1 - a, \quad R = \begin{pmatrix} a & -b \\ b & a \end{pmatrix}. \tag{2}$$

For arbitrary $z = (u, v) \in \mathbb{R}^2$ and $e = (1, 0)$, set

$$c_i = R^i e, \quad d_i = R^i z, \quad C = \operatorname{conv}\{c_i : 0 \leq i < m\}, \quad D = \operatorname{conv}\{d_i : 0 \leq i < m\}. \tag{3}$$

Indices are modulo m. The demanded transitions are exactly $(c_i, d_i) \mapsto (c_{i+1}, d_{i+1})$. Thus the two polygons have the same angular spacing and are paired in the same cyclic order; other pairings and other polygon shapes are outside the class.

2 Complete classification

Theorem 1. *For the class* (2)–(3)*, all demanded transitions of* (1) *hold if and only if one of the following alternatives holds.*

1. $\theta = 0$, $m \geq 5$, *and*

$$r = 2\delta = 4\sin^2(\pi/m), \qquad u = \frac{1}{2}, \qquad v = -\frac{b}{2\delta}. \tag{4}$$

2. $\theta = 0$, $m = 4$, *and, for some real* w,

$$u = \frac{w}{2}, \qquad v = -\frac{w}{2}, \qquad 0 < w \leq 1, \qquad rw \geq 2. \tag{5}$$

In particular, for every $\theta > 0$ *the feasible set is empty. At* $\theta = 0$ *the smallest feasible product for each fixed* $m \geq 4$ *is* $4\sin^2(\pi/m)$*. For the square the feasible products are all* $r \geq 2$*, with the unique minimizing choice* $w = 1$*.*

Every feasible case has a saddle/KKT pair $(0, 0)$*. Initializing at* $(x_0, y_0) = (c_{m-1}, d_{m-1})$ *yields an exact nonconstant orbit of minimal period* m*.*

The square branch can equivalently be written $r \geq 2$ and $2/r \leq w \leq 1$. For any fixed allowed w, it contains the full half-line $r \geq 2/w$. For $m \geq 5$, by contrast, (4) is the only allowed product and dual vertex, not merely an optimizer.

2.1 Projection criterion with closed normal cones

For a nonempty closed convex set E, write

$$N_E(v) = \{n : \langle n, q - v\rangle \leq 0 \text{ for all } q \in E\}, \qquad v \in E.$$

The exact Euclidean projection test is

$$v = P_E(s) \quad \Longleftrightarrow \quad v \in E, \quad s - v \in N_E(v). \tag{6}$$

For necessity, compare the squared distances to v and $v + \varepsilon(q - v)$ and let $\varepsilon \downarrow 0$. For sufficiency, expand

$$\|s - q\|^2 = \|s - v\|^2 + \|q - v\|^2 - 2\langle s - v, q - v\rangle.$$

This also proves uniqueness: if $q \neq v$, the right side is strictly greater than $\|s - v\|^2$. On a convex hull, testing its vertices suffices by affinity. The inequalities are non-strict; normal vectors on cone boundaries are fully admissible.

3 Every vertex test and its adjacent reduction

Because $RC = C$ and $RD = D$, Euclidean projections commute with R. It suffices to analyze the phase $m - 1 \to 0$. Its actual residuals are

$$p = R^{-1}e - rR^{-1}z - e, \qquad h = R^{-1}z - z + (1+\theta)e - \theta R^{-1}e. \tag{7}$$

Define, for every $k = 0, \dots, m-1$,

$$P_k = \langle p, R^k e - e \rangle, \qquad D_k = \langle h, R^k z - z \rangle.$$

The demanded projections are equivalent to $P_k, D_k \le 0$ for every k. Put $P_+ = P_1$, $P_- = P_{m-1}$, and similarly for $D_\pm$.

For $1 \le k < m$, let $\beta = k\alpha$ and set

$$A_k = \frac{1-\cos\beta}{2\delta} + \frac{\sin\beta}{2b}, \qquad B_k = \frac{1-\cos\beta}{2\delta} - \frac{\sin\beta}{2b}.$$

Then a componentwise identity gives

$$R^k e - e = A_k(Re - e) + B_k(R^{-1}e - e). \tag{8}$$

Both coefficients are nonnegative. Indeed, $\beta/2 \in [\alpha/2, \pi - \alpha/2]$, and monotonicity and symmetry of cotangent imply

$$\left|\cot\frac{\beta}{2}\right| \le \cot\frac{\alpha}{2} = \frac{b}{\delta}, \qquad \frac{|\sin\beta|}{b} \le \frac{1-\cos\beta}{\delta}.$$

The linear map $T_z = \left(\begin{smallmatrix} u & -v \\ v & u \end{smallmatrix}\right)$ commutes with R and sends e to z. Applying it to (8) proves the same identity for dual edges, even when $z = 0$. Therefore

$$P_k = A_k P_+ + B_k P_-, \qquad D_k = A_k D_+ + B_k D_-. \tag{9}$$

The $k = 0$ tests vanish. Hence the four adjacent inequalities are necessary and sufficient for all vertices at all phases.

3.1 Exact scalar identities

We have $0 \le a < 1$, $b > 0$, $\delta > 0$, and $b^2 = \delta(1+a)$. Introduce the invertible coordinates

$$q = \frac{b}{\delta}, \quad w = u - qv, \quad j = u + \frac{v}{q}, \qquad u = \frac{\delta w + (1+a)j}{2}, \quad v = \frac{b(j-w)}{2}. \tag{10}$$

Writing $N = u^2 + v^2$, expansion of (7) gives

$$\begin{aligned} P_- &= \delta(2 - rw), \\ P_+ &= \delta\{r[aw + (1+a)j] - 2a\}, \\ D_- &= \delta(2N - 2\theta u - w), \\ D_+ &= -aD_- - \delta(1+a)\{(1+\theta\delta)j - \theta\delta w\}. \end{aligned} \tag{11}$$

These identities use the unscaled primal residual in (7).

4 Proof of the parameter classification

Assume first that all four expressions in (11) are nonpositive. The first primal inequality gives

$$rw \geq 2, \qquad w > 0.$$

The second primal inequality, together with $a \geq 0$, yields

$$r(1+a)j \leq 2a - raw = a(2 - rw) \leq 0, \qquad j \leq 0. \tag{12}$$

Since $D_+ \leq 0$, $D_- \leq 0$, and $a \geq 0$, the last identity in (11) also gives

$$0 \geq D_+ + aD_- = -\delta(1+a)\{(1+\theta\delta)j - \theta\delta w\}. \tag{13}$$

Consequently $(1+\theta\delta)j - \theta\delta w \geq 0$. If $\theta > 0$, its left side is strictly negative because $j \leq 0$, $w > 0$ and $\delta > 0$. This contradiction excludes every positive-extrapolation case, without dividing by a or assuming any strict normal inequality.

4.1 Zero extrapolation and polygons with at least five vertices

For $\theta = 0$, (12)–(13) force $j = 0$. The inverse coordinates and the norm become

$$u = \frac{\delta w}{2}, \qquad v = -\frac{bw}{2}, \qquad N = \frac{\delta w^2}{2}.$$

Substitution reduces the four tests to

$$\begin{aligned} P_- &= \delta(2 - rw), & P_+ &= -aP_-, \\ D_- &= \delta w(\delta w - 1), & D_+ &= -aD_-. \end{aligned} \tag{14}$$

If $m \geq 5$, then $a > 0$. Both opposite-sign pairs can be nonpositive only when $P_- = D_- = 0$. Thus $rw = 2$ and $\delta w = 1$, proving (4). Conversely, (4) makes every expression in (14) zero, so (9) and (6) prove sufficiency. In fact $p = h = 0$: the unprojected inputs already equal the target vertices.

4.2 The exceptional square family

For $m = 4$, $a = 0$ and $b = \delta = 1$. The plus tests in (14) vanish identically. The remaining inequalities hold exactly when $rw \geq 2$ and $0 < w \leq 1$, giving (5). Every such choice is sufficient by (9). The smallest possible r is 2, and $r = 2$ forces $w = 1$. Every $r \geq 2$ is feasible, for example with $w = 1$.

Finally, $2\delta = 2(1 - \cos(2\pi/m)) = 4\sin^2(\pi/m)$. This proves all parameter and fixed-m minimum assertions of Theorem 1.

5 Boundary normals and the actual initialized orbit

For the square branch, the residuals at phase $m-1 \to 0$ are explicitly

$$p = \left(\frac{rw}{2} - 1\right)(1,1), \qquad h = (1-w, 0). \tag{15}$$

The primal normal lies on a boundary ray of $N_C(e)$. To express the dual normal in the same base coordinates, its pullback is

$$T_z^{\mathsf{T}} h = \frac{w(1-w)}{2}(1,1).$$

This is a transformed normal, not the actual dual residual h. The primal residual vanishes exactly when $rw = 2$, and the dual residual vanishes exactly when $w = 1$. At $r = 2$, both vanish. For $m \geq 5$, both residuals vanish throughout the feasible class.

When the square primal residual is nonzero, the associated supporting face is the edge from c_0 to c_1. When its dual residual is nonzero, that face is the edge from d_0 to d_1. These equalities in the normal tests do not create multiple Euclidean projections: the strictly positive $\|q - v\|^2$ term in the proof of (6) still excludes every other projected point. At zero residual all vertex normal tests are equalities. The limiting choice $w = 0$ is excluded by $rw \geq 2$, so no collapsed dual polygon has been silently added.

5.1 Initialization and least period

Use precisely the current state

$$(x_0, y_0) = (c_{m-1}, d_{m-1}). \tag{16}$$

The verified primal projection produces c_0. The dual step uses this newly produced c_0 and then produces d_0. Rotational equivariance and induction give

$$(x_n, y_n) = (c_{n-1 \bmod m}, d_{n-1 \bmod m}), \qquad n \geq 0.$$

The m primal vertices are distinct, so the least period is exactly m. Every feasible dual vertex has $u > 0$, so D is also a nondegenerate regular polygon. The orbit is nonconstant and does not converge, also not weakly in this finite-dimensional space.

5.2 Convexity, saddle point, and normalized product

Both polygons are nonempty compact convex sets, so their indicators are proper closed convex functions. Their vertex sums vanish, and therefore $0 \in C \cap D$ (indeed, in both interiors for feasible parameters). The inclusions

$$-L^*0 = 0 \in N_C(0), \qquad L0 = 0 \in N_D(0)$$

prove that $(0,0)$ is a KKT pair, equivalently a saddle of $f(x) + \langle x, y\rangle - g^*(y)$. The coupling is the identity on the ordinary Euclidean plane, so $\|L\| = 1$ and $\tau\sigma\|L\|^2 = r$ exactly. This completes Theorem 1.

6 Infima and relation to the triangle constructions

At $\theta = 0$, the minima satisfy

$$0 < 4\sin^2(\pi/m) \le 2 < 3 < 4 \qquad (m \ge 4).$$

They strictly decrease to zero as $m \to \infty$. Thus the infimum over these finite polygons is zero and is not attained. This does not construct an orbit at $r = 0$. For $\theta > 0$, the present class is empty; its infimum may be written $+\infty$ only under the extended-real convention for an empty feasible set.

The separately established ordered triangle family [3] has minimum

$$r_3(\theta) = \frac{2\sqrt{1 + 3\theta + 3\theta^2} + 1 + 3\theta}{(1+\theta)^2}.$$

At zero extrapolation this equals 3. Its positive-extrapolation examples, including the $\theta = 1/8$ setting of the earlier construction [2], are not contradicted by Theorem 1: for a triangle, $a = \cos(2\pi/3) < 0$, so the sign argument above does not apply.

For completeness, the bilinear two-cycle benchmark is $r_2(\theta) = 4/(1 + 2\theta)$. It follows directly from the scalar unrestricted map: demanding $(x, y) \mapsto (-x, -y)$ gives $ry = 2x$ and $2y = (1 + 2\theta)x$. Comparing it with the triangle formula gives

$$r_3(\theta) < r_2(\theta) \quad \Longleftrightarrow \quad 0 \le \theta < \frac{4\sqrt{5} - 5}{22}.$$

Indeed, writing $W = 1 + 3\theta + 3\theta^2$ and rationalizing $r_3 = 3/(2\sqrt{W} - 1 - 3\theta)$ reduces the comparison to $7 + 18\theta < 8\sqrt{W}$. Both sides are positive, and

$$(7 + 18\theta)^2 - 64W = 3(44\theta^2 + 20\theta - 5).$$

At $\theta = 1/4$ the triangle product is $(28 + 8\sqrt{31})/25 > 8/3$, while $r_2 = 8/3$. The regular polygons here supply no positive-extrapolation improvement at that parameter, or at any other.

6.1 Scope of the conclusions

The nonexistence result excludes only the prescribed paired regular polygons in the prescribed cyclic order, with $L = I$, $\sigma = 1$, and the literal map (1). It does not exclude differently paired polygons, nonregular polygons, other geometries, or arbitrary cycles. It does not establish a universal convergence threshold or universal sharpness. The construction also makes no claim of attraction or robustness. Each feasible cycle is bounded, and its ordinary Cesàro averages converge to $(0, 0)$ because every full block of m states sums to zero.

7 Exact symbolic reproducibility

The source package includes `verify.py`, an exact symbolic certificate using Python 3 and SymPy. Running `python3 verify.py` checks the following identities directly from the update residuals:

1. all four adjacent normal expressions (11) and the inverse change of coordinates (10);
2. the primal and dual edge decompositions (8), hence both all-vertex normal identities (9);
3. all four zero-extrapolation reductions (14), the unique $m \geq 5$ attainment, and the vanishing of both residuals there;
4. the entire square branch, its residual formulas (15), and the transformed dual normal;
5. the algebraic sign decomposition behind the positive-extrapolation contradiction and the polynomial used for the triangle comparison.

The verifier reduces rational identities modulo $b^2 = 1 - a^2$ after clearing denominators. The denominators are nonzero in the geometric domain $0 \leq a < 1$, $b > 0$. No floating-point arithmetic, parameter grid, numerical optimization, or finite sampling is used as evidence for the classification.

The proof supplies the trigonometric signs, the nonnegativity of A_k, B_k, the projection criterion, and all quantifiers. The script certifies the algebra. The separate triangle theorem is used only in Section 6.

Run `bash build.sh` to compile twice with pdfLaTeX. The clean source package contains the manuscript, the unchanged style from report 18, the build script, the verifier, its result, and a README.

References

[1] A. Chambolle and T. Pock, A first-order primal-dual algorithm for convex problems with applications to imaging, *Journal of Mathematical Imaging and Vision* **40** (2011), 120–145. doi:10.1007/s10851-010-0251-1.

[2] *An exact period-three Chambolle–Pock orbit at $\theta = 1/8$ below the quadratic threshold.* Companion mathematical report 17, 6 October 2026.

[3] *Exact planar Chambolle–Pock three-cycles and an optimal ordered triangle family.* Companion mathematical report 18, 6 October 2026.